

5.13 Classwork: Exponential Differentiation — Markscheme

1. [Maximum mark: 4]

The function g is defined by $g(x) = e^{x^2+1}$, where $x \in \mathbb{R}$. Find $g'(-1)$.

Solution:

Differentiate using the chain rule:

$$g'(x) = e^{x^2+1} \cdot 2x \quad (\text{M1})(\text{A1})$$

Substitute $x = -1$:

$$g'(-1) = e^{(-1)^2+1} \cdot 2(-1) = e^2 \cdot (-2) \quad (\text{A1})$$

$$g'(-1) = -2e^2 \quad (\text{A1})$$

2. [Maximum mark: 16]

Let $y = \frac{\ln x}{x^4}$ for $x > 0$.

(a) Show that $\frac{dy}{dx} = \frac{1 - 4 \ln x}{x^5}$. [3]

Solution:

Using the quotient rule with $u = \ln x$, $v = x^4$:

$$\frac{dy}{dx} = \frac{\frac{1}{x} \cdot x^4 - \ln x \cdot 4x^3}{(x^4)^2} \quad (\text{M1})$$

$$= \frac{x^3 - 4x^3 \ln x}{x^8} \quad (\text{A1})$$

$$= \frac{x^3(1 - 4 \ln x)}{x^8} = \frac{1 - 4 \ln x}{x^5} \quad (\text{A1})(\text{AG})$$

(b) The graph of f has a horizontal tangent at point P . Find the coordinates of P . [5]

Solution:

$$\text{Horizontal tangent: } f'(x) = 0 \quad (\text{M1})$$

$$\frac{1 - 4 \ln x}{x^5} = 0$$

$$\text{Since } x^5 > 0 \text{ for } x > 0: \quad 1 - 4 \ln x = 0 \quad (\text{A1})$$

$$\ln x = \frac{1}{4} \implies x = e^{1/4} \quad (\text{A1})$$

$$\text{Substitute into } f(x): \quad (\text{M1})$$

$$y = \frac{\ln(e^{1/4})}{(e^{1/4})^4} = \frac{\frac{1}{4}}{e} = \frac{1}{4e}$$

$$P = \left(e^{1/4}, \frac{1}{4e} \right) \quad (\text{A1})$$

(c) Given that $f''(x) = \frac{20 \ln x - 9}{x^6}$, show that P is a local maximum point. [3]

Solution:

$$\text{Substitute } x = e^{1/4} \text{ into } f''(x): \quad \ln(e^{1/4}) = \frac{1}{4} \quad (\text{M1})$$

$$f''(e^{1/4}) = \frac{20 \cdot \frac{1}{4} - 9}{(e^{1/4})^6} = \frac{5 - 9}{e^{3/2}} = \frac{-4}{e^{3/2}} \quad (\text{A1})$$

$$\text{Since } \frac{-4}{e^{3/2}} < 0, P \text{ is a local maximum point.} \quad (\text{R1})(\text{AG})$$

(d) Solve $f(x) > 0$ for $x > 0$. [2]

Solution:

$$\frac{\ln x}{x^4} > 0$$

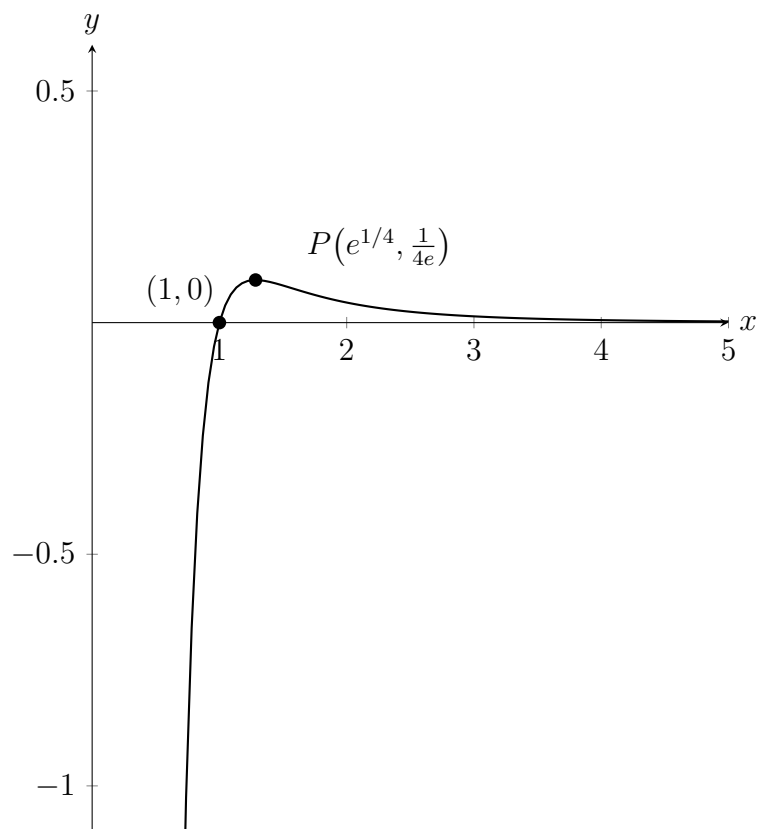
$$\text{Since } x^4 > 0 \text{ for all } x > 0, \text{ the sign depends on } \ln x. \quad (\text{R1})$$

$$\ln x > 0 \iff x > 1$$

$$\text{Answer: } x > 1 \quad (\text{A1})$$

(e) Sketch the graph of f , showing clearly the value of the x -intercept and the approximate position of point P . [3]

Solution:



(A1) for correct general shape (increasing from negative, crossing axis, reaching max, decreasing toward 0)

(A1) for x -intercept at $x = 1$

(A1) for point P approximately positioned (just right of $x = 1$, small positive y)